

Programming Fundamentals

C Programming Language

Assignment No. 3

if Statement

1. Write a C program that takes an integer from the user and uses an if statement to check whether the number is positive.
2. Write a C program that takes the age of a person and displays "Eligible to vote" if the age is 18 or above.
3. Write a C program that takes an integer and uses an if statement to check whether it is divisible by 5.
4. Write a C program that takes marks obtained by a student and displays "Pass" if the marks are 50 or above.

if-else Statement

1. Write a C program that takes an integer and determines whether the number is even or odd using if-else.
2. Write a C program that takes an integer and determines whether it is positive or negative using if-else.
3. Write a C program that takes two integers and displays the larger number using if-else.
4. Write a C program that takes a person's age and determines whether the person is eligible for a driving license. Assume the minimum age is 18.

else-if Statement

1. Write a C program that takes marks from 0 to 100 and displays the grade according to the following criteria:
80–100 → A
70–79 → B
60–69 → C
50–59 → D
Below 50 → F
2. Write a C program that takes three integers and determines the largest number using an else-if structure.
3. Write a C program that takes temperature in Celsius and displays:
40 or above → Very Hot
30–39 → Hot
20–29 → Normal
10–19 → Cold
Below 10 → Very Cold
4. Write a C program that takes the number of electricity units consumed and calculates the bill according to these rates:
0–100 units → Rs. 10 per unit
101–200 units → Rs. 15 per unit
201–300 units → Rs. 20 per unit
Above 300 units → Rs. 25 per unit
5. Write a C program that takes marks and displays the student's performance:
90–100 → Excellent
80–89 → Very Good
70–79 → Good
60–69 → Satisfactory
Below 60 → Needs Improvement

Nested if Statements

1. Write a C program that asks the user to enter a username and password. Use nested if statements to verify:
Username is correct.
If the username is correct, check whether the password is correct.
Display an appropriate message for each case.
2. Write a C program that takes a student's marks and family income. A student is eligible for a scholarship if:
Marks are 80 or above, and
Family income is less than or equal to Rs. 50,000.
3. Write a C program that takes three numbers and uses nested if statements to find the smallest number.

Switch Statement

1. Write a C program that takes two numbers and an operator (+, -, *, /) and performs the selected arithmetic operation using a switch statement.
2. Write a C program that takes a number from 1 to 7 and displays the corresponding day:
1 → Monday
2 → Tuesday
3 → Wednesday
4 → Thursday
5 → Friday
6 → Saturday
7 → Sunday
3. Write a C program that displays the following menu using a switch statement:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Ask the user to select an option and perform the corresponding operation.
4. Write a C program that takes a month number from 1 to 12 and uses a switch statement to display the number of days in that month. Assume February has 28 days.

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